Telecom Customer Churn Analytics Report

Project Domain: Telecom Analytics

Project Type: End-to-End Data Analytics Project

Technologies Used:
Python | Machine Learning | SQL | Excel | Power BI

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Technical Stack

Programming & Data Processing:
- **Python –** Used for data cleaning, exploratory data analysis (EDA), feature engineering, and machine learning model development

Database:
- **MySQL –** Used for structured data storage, churn pattern analysis, and validation using joins and aggregations

Visualization & Reporting:
- **Power BI –** Built an interactive multi-page dashboard with KPIs, churn insights, and segmentation analysis using DAX and Power Query
- **Microsoft Excel –** Used for pivot-based analysis, KPI validation, and intermediate business reporting

Libraries & Frameworks:
- **Pandas –** Data manipulation and transformation
- **NumPy –** Numerical computations
- **Matplotlib & Seaborn –** Exploratory data visualization

• **Scikit-learn –** Model building, evaluation, and preprocessing
• **XGBoost –** Advanced gradient boosting model for improved prediction accuracy

Machine Learning Models:
• **Logistic Regression –** Baseline model for churn classification
• **Random Forest Classifier** – Ensemble model for improved performance and feature importance analysis
• **XGBoost Classifier –** Optimized model for high-accuracy churn prediction

Project Overview

Customer churn is a critical challenge in the telecom industry, directly impacting revenue and long-term customer retention. This project, *ChurnX Insights — Telecom Customer Analytics Report*, delivers an end-to-end analytics solution combining descriptive and predictive techniques to understand and reduce churn.

Python was used for data cleaning, exploratory data analysis (EDA), feature engineering, and machine learning model development. MySQL and Excel supported data validation and business-level analysis, while Power BI was used to build an interactive, multi-page dashboard for stakeholder decision-making.

The project integrates historical trend analysis with machine learning–based churn prediction to identify high-risk customers and uncover key behavioral patterns. The final solution provides actionable insights that enable telecom businesses to design targeted retention strategies and improve customer lifetime value.

Business Problem Statement

Telecom companies experience significant customer churn driven by competitive pricing, service quality issues, and flexible contract options. Identifying customers at risk of leaving before churn occurs is a major business challenge.

Traditional reporting methods offer only historical insights and lack predictive capability, limiting proactive decision-making. This project addresses the need for a data-driven approach to analyze churn drivers and forecast customer churn.

By leveraging machine learning models and analytical techniques, the solution enables early identification of high-risk customers, helping businesses reduce churn, enhance retention strategies, and optimize revenue performance.

Project Objectives

• Analyze customer behavior to identify key drivers influencing churn
• Perform exploratory data analysis (EDA) to uncover patterns across tenure, contract type, services, and payment methods

• Develop and evaluate machine learning models to predict churn probability
• Segment customers into risk categories to identify high-risk churn groups
• Validate analytical findings using SQL queries and Excel-based analysis
• Design an interactive Power BI dashboard to deliver insights for business decision-making

Dataset Description

The dataset used in this project is a publicly available Telecom Customer Churn dataset, widely used for churn analysis and predictive modeling. It contains approximately 7,000 customer records with a mix of categorical and numerical features.

The dataset includes customer demographics, service subscriptions, contract details, and billing information. The target variable, *Churn*, indicates whether a customer has discontinued the service.

Key attributes include:

• Customer Details – Gender, Senior Citizen status, tenure

• Service Information – Internet service, phone service, add-on services

• Contract & Billing – Contract type, payment method, monthly charges, total charges

• Target Variable – Churn (Yes/No)

This dataset provides a comprehensive view of customer behavior and supports both descriptive analysis and predictive modeling.

Exploratory Data Analysis (EDA)

EDA was performed to identify patterns and trends associated with customer churn. The analysis revealed a significant proportion of customers discontinuing services.

Key findings from EDA include:

• Customers on month-to-month contracts exhibit higher churn rates

• Fiber optic internet users show increased churn compared to other service types

• Customers with shorter tenure are more likely to churn than long-term subscribers. • Payment method and service combinations influence churn behavior

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These insights helped identify critical churn drivers, guided feature selection for machine learning models, and informed dashboard design.

Machine Learning Methodology

The dataset was split into training and testing sets to evaluate model performance. Data preprocessing included encoding categorical variables and scaling numerical features where required.

Multiple classification models were developed:

• Logistic Regression – Baseline model for churn prediction

• Random Forest – Ensemble model capturing non-linear relationships

• XGBoost – Optimized gradient boosting model for improved accuracy

Models were evaluated using appropriate performance metrics to ensure reliable comparison and selection of the best-performing approach. This enabled accurate identification of high-risk churn customers.

SQL Analysis

MySQL was used to perform structured analysis and validate churn insights generated through Python and machine learning workflows. The dataset was stored in a relational database to enable efficient querying and aggregation.

Key SQL analyses included:

• Churn distribution across contract types, tenure, and services

• Payment method–based churn segmentation

• Identification of high-risk customer groups

SQL served as a validation layer, ensuring consistency and reliability of analytical results across tools.

Excel Analysis & Validation

Microsoft Excel was used for business-level validation and interpretation of churn insights. Pivot tables and charts were created to analyze churn across key dimensions such as contract type, tenure, payment method, and demographics.

KPI summaries and visual checks were performed to validate trends identified through SQL and Python. This step improved confidence in the analysis and ensured alignment between technical outputs and business understanding.

Excel acted as an intermediate validation layer before final visualization in Power BI.

Power BI Dashboard Overview

Power BI was used to develop an interactive, multi-page dashboard that presents telecom churn insights in a clear, business-oriented format. The dashboard integrates cleaned data, engineered features, and machine learning outputs to enable data-driven decision-making.

A dark-themed interface was implemented to improve visual clarity and user experience. DAX measures were created to compute key KPIs such as churn rate, average tenure, average monthly charges, and high-risk customer percentage. Interactive slicers allow dynamic filtering across demographics, services, contract types, and payment methods.

The dashboard acts as a centralized analytics solution, transforming complex churn data into actionable business insights and supporting both descriptive and predictive analysis.

Dashboard Structure

The Power BI dashboard is designed as a **five-page interactive report**, with each page focusing on a specific business perspective of customer churn.

1. Executive Summary
Provides a high-level overview of key KPIs including total customers, churn rate, average tenure, and high-risk customer percentage. Designed for quick decision-making with interactive filters for rapid insight generation.

2.Customer Demographics Analysis
Analyzes churn patterns across gender, senior citizen status, partner/dependent relationships, and tenure groups. Identifies customer segments with higher churn probability.

3.Services & Usage Behavior
Examines churn trends based on subscribed services such as internet type, tech support, streaming, and security features. Highlights service combinations that significantly impact churn risk.

4.Billing & Payment Insights
Evaluates the relationship between churn and billing factors including payment methods, monthly charges, and total charges. Identifies pricing and payment-related churn drivers.

5.Churn Risk Prediction (ML Insights)
Presents machine learning–based churn predictions, including high-risk customer segmentation and probability scores. This page bridges analytics with business action by identifying customers who require targeted retention strategies.

Key Insights

• Customers on month-to-month contracts exhibit the highest churn rates, indicating low customer commitment and higher volatility
• Fiber optic internet users show increased churn compared to DSL and non-internet users, suggesting potential service or pricing concerns
• Customers with shorter tenure are significantly more likely to churn than long-term subscribers
• Electronic check payment method is strongly associated with higher churn, while automated payment options show better retention
• Higher monthly charges correlate with increased churn risk, indicating price sensitivity among customers
• Machine learning models identified a substantial segment of high-risk customers, enabling proactive churn prediction and intervention

Business Recommendations

- Encourage migration from month-to-month to long-term contracts through discounts, bundled offers, and loyalty programs
- Improve service quality and pricing strategies for fiber optic users to reduce dissatisfaction and churn Implement early engagement programs for new customers to reduce churn during the initial tenure period.
- Implement early-stage engagement and onboarding programs to reduce churn during initial customer tenure Introduce personalized retention offers for customers identified as high-risk through machine learning predictions.
- Promote automated payment methods by offering incentives to customers

Limitations

- The dataset represents a historical snapshot and does not capture real-time customer behavior
- Key factors such as customer satisfaction, network quality, and competitor offerings are not included
- Model performance is dependent on the quality and scope of available features
- Class imbalance in churn data may affect prediction accuracy for churned customers
- The analysis assumes stable customer behavior patterns, which may vary over time

Future Scope

- Integrate real-time data pipelines to enable live churn monitoring and automated alerts
- Incorporate additional features including customer feedback, network performance, and usage behavior
- Automate churn risk alerts and integrate insights with CRM systems for proactive retention strategies
- Deploy the churn prediction model as a production-ready solution for continuous business use
- Improve prediction accuracy by experimenting with advanced models such as deep learning and ensemble techniques

Conclusion

This project delivers a comprehensive end-to-end telecom churn analytics solution by combining data analysis, machine learning, and business intelligence techniques. Through exploratory analysis and predictive modeling, key churn drivers and high-risk customer segments were effectively identified.

The Power BI dashboard translates complex analytical outputs into clear, actionable insights, enabling data-driven decision-making. The project highlights strong capabilities in data processing, model development, and visualization, while providing practical business value in improving customer retention and optimizing revenue strategies.